# RESEARCH 1

## ACADEMIC

# Grade 11/12

## Curriculum Guide

## RESEARCH 1
***Prerequisite: None***

### Course Description

This course develops the learners' critical thinking, argumentation, and problem-solving skills through the research process. It covers the nature and ethics of research, identification of research problems, and the review of related literature to construct evidence-based arguments. The learners formulate research questions, justify problems using credible sources, and design appropriate research methodologies.

The course also introduces user-centered analysis and explores possible solution directions, preparing the learners to develop a proposal that may serve as a foundation for further research or innovation projects.

**Elective:** Academic
**Time Allotment:** 80 hours for one term

| CONTENT DOMAIN | CONTENT STANDARDS<br>*The learners demonstrate understanding of …* | LEARNING COMPETENCIES<br>*The learners …* | SUGGESTED OUTPUT |
|---|---|---|---|
| **Foundations of Research** | 1. the nature, purposes, and processes of research. | 1. explain the purpose of research in solving real-world problems.<br>2. differentiate the various types of research, including basic, applied, action, and evaluation research.<br>3. describe the basic steps in the research process.<br>4. recognize characteristics of a good research problem.<br>5. analyze real-world problems using a user-centered or needs-based approach.<br>6. propose possible solution directions based on the problem. | • Reflection: "Why this problem matters"<br>• Examples of strong and weak problems<br>• Solution justification using evidence |

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|---|---|---|---|
| | | 7. construct arguments supporting the feasibility and relevance of proposed solutions.<br>8. examine feasibility considering ethical, social, and environmental factors. | |
| **Research Ethics** | 2. ethical standards and responsible conduct in research. | 9. apply ethical standards in all stages of the research process.<br>10. demonstrate proper citation of sources in writing the research paper. | • Properly cited written outputs<br>• Ethics checklist, including informed consent procedures |
| **Review of Literature for Identifying Research Problems** | 3. how reviewing literature helps identify research gaps, issues, and questions that may lead to the formulation of research problems. | 11. evaluate the credibility and relevance of sources.<br>12. synthesize related literature to identify gaps, issues, or needs.<br>13. formulate a clear and researchable problem statement.<br>14. develop aligned research questions or hypotheses.<br>15. construct a clear, evidence-based argument to justify the research problem using the claim–evidence–reasoning (CER) framework.<br>16. identify basic assumptions and limitations of the study. | • Annotated sources<br>• Literature synthesis matrix<br>• CER matrix<br>• Problem justification write-up<br>• Structured research introduction<br>• Identification of research gaps leading to the proposed problem<br>• General and specific research questions |
| **Research Design and Methodology** | 4. appropriate research designs and methodological procedures. | 17. select an appropriate research design based on the problem.<br>18. identify key variables or focus of the study.<br>19. determine appropriate participants or data sources.<br>20. develop data collection methods and basic research instruments.<br>21. outline the research procedure. | • Research design justification<br>• Draft instruments<br>• Procedure flowchart<br>• Stakeholder/needs analysis |

| CONTENT DOMAIN | CONTENT STANDARDS<br><br>*The learners demonstrate understanding of …* | LEARNING COMPETENCIES<br><br>*The learners …* | SUGGESTED OUTPUT |
|---|---|---|---|
| | | 22. justify the alignment of methodology with the research problem. | • Methodology justification write-up<br>• Feasibility analysis |
| **Research Proposal Development** | 5. the structure and presentation of a research proposal. | 23. organize a research proposal using a clear argumentative structure (problem–evidence–method–solution).<br>24. prepare a comprehensive research proposal.<br>25. present and defend the research proposal using evidence-based reasoning.<br>26. revise the proposal based on feedback and counter-arguments. | • Complete research proposal<br>• Argument-driven presentation slides<br>• Oral defense<br>• Revised proposal |

**PERFORMANCE STANDARDS:**
The learners can develop and present a comprehensive, well-structured research proposal that clearly justifies a research problem through evidence-based argumentation, applies appropriate research design and methodology, and may serve as a foundation for further research or innovation projects while adhering to ethical standards.